



ADAMJEE GOVT. SCIENCE COLLEGE

PRELIMINARY EXAMINATION - 2026

Time: 20 Minutes

PHYSICS – II

Max. Marks: 17

SECTION “A”

(Multiple choice questions)

NOTE: i) This section consists of 17 part questions and all are to be answered. Each question carries one mark.
ii) Don't copy down the part questions in your answer book. Write only the answer in full against the proper number of the question and its part.

Q1. Choose the correct answer for each from the given options:

- i) Of the following, Compton scattering from electrons is most easily observed for:
 *Infrared light *Visible light *Ultraviolet light *X-rays
- ii) The primary purpose of a truth table in digital electronics is to:
 *Determine input voltage *Measure circuit resistance
 *Analyze output states based on input combinations *Calculate circuit power
- iii) Eddy currents produce:
 *Static electricity in conductive materials *Alternating magnetic fields with no heating effects
 *Magnetic fields that oppose the inducing field and cause heating in conductors
 *Uniform electric fields that have no impact on power generation
- iv) In alternating current, the flow of electric charge periodically reverses direction. The term for this specific phenomenon is:
 *Alternation *Oscillation *Cycle *Resistance
- v) Pressure and volume in an adiabatic process are related as:
 * $PV = \text{constant}$ * $PV^\gamma = \text{constant}$ * $P^\gamma V = \text{constant}$ * $(PV)^\gamma = \text{constant}$
- vi) The half-life of a sample of radioactive element of 16 kg is 2 years what will be the remaining amount after 6 years
 *2 kg *4 kg *6 kg *8 kg
- vii) An inductor stores electric potential energy in:
 *The electric field around it *The magnetic field within its coil
 *The capacitance of its windings *The heat generated by its resistance
- viii) Which device is not used in a diesel engine?
 *Outlet Valve *Piston *Sparking Plug *Injector
- ix) The reciprocal of Bulk Modulus is called:
 *Compressibility *Volume Stress *Modulus of Rigidity *Volume Strain
- x) The volume of a given mass of a gas will be doubled at atmospheric pressure if the temperature of the gas is changed from 40°C to:
 * 300°C * 353°C * 423°C * 80°C
- xi) The low temperature reservoir of heat of a Carnot engine is at 7°C and has an efficiency of 40%. It is desired to increase the efficiency to 50%. By how much degree the temperature of hot reservoir is increased?
 *93.4K *363K *9.34K *9293K
- xii) What is the self-inductance of a coil which produces 5 V when the current changes from 3 to 2A in 1ms?
 *5000 Henry *5 milli-Henry *50 Henry *5 Henry
- xiii) Donor impurities are:
 *Ge & Si *Ga & In *As & Sb *None of these
- xiv) What shunt resistance must be connected across a galvanometer of 50.0Ω resistance which gives full scale deflection with 2.0 mA current, so as to convert it into an ammeter of 10.0 A?
 * 10.2Ω * 12.5Ω * 0.01Ω * 9.5Ω
- xv) K_β characteristic X-rays are produced due to transition of electrons
 *From L-shell of K-shell *From M-shell to L-shell *From M-shell to K-shell *From N-shell to K-shell
- xvi) In the equation ${}^{27}_{13}\text{Al} + {}^4_2\text{He} \rightarrow {}^{30}_{15}\text{P} + \text{X}$, The correct symbol for X is:
 * ${}^1_0\text{e}$ * ${}^1_1\text{H}$ * ${}^4_2\text{He}$ * ${}^0_0\text{n}$
- xvii) A refrigerator with its door open. The temperature of the room will:
 *Rise *Fall
 *Remains the same *Rise or fall depending on the area of the room

SECTION “B”

(SHORT ANSWER QUESTIONS)

(36 Marks)

NOTE: Attempt any Nine part questions from this section. All questions carry equal marks.

Q2.

- i. A 50 g piece of copper at 100°C is placed in 200 g of water at 20°C . If the final temperature of the system is 21.8°C , calculate the specific heat capacity of copper. (Specific heat capacity of water = $4.18 \text{ J/g}^{\circ}\text{C}$) OR Compute the magnitude of the magnetic field of a long, straight wire carrying a current of 1 A at distance of 1m from it. Compare it with Earth's magnetic field which is $50 \mu\text{T}$.
- ii. A heat engine performs work of 0.4166 watts in one hour and rejects 4500J of heat to the sink. What is the efficiency of engine? OR Discuss energy stored in an inductor. Derive its relevant formula.
- iii. Discuss AND gate OR gate with variable input truth table.
- iv. At what relative velocity the mass of particle becomes doubled. OR How fast would a rocket have to go relative to an observer for its length to be contracted to 99% of its length at rest?
- v. State Ampere's Law, Lenz's Law, Biot and Savart's Law and Wien's displacement law. OR Prove $C_p - C_v = R$.
- vi. A 500 turn coil in an A.C generator having an area of 1000 cm^2 rotates in a magnetic field of value 50 Tesla. In order to generate 220 V Max. How fast is the coil to be rotated? Express your answer in terms of the number of revolutions per seconds. OR Write down the construction and working of G.M Counter OR WILSON CLOUD CHAMBER.
- vii. What is galvanometer? Explain the conversion of galvanometer into ammeter and voltmeter.
- viii. A coil with an inductance of 0.5 H experiences a rate of change of current of 2 A/s. Calculate the induced EMF in the coil. OR An inductor have reactance of 500Ω is connected with alternating voltage 220V and frequency of 50 Hz.
- ix. Write four difference between petrol and diesel engine. OR Explain the process of pair production briefly.
- x. What is half-life of a radioactive element? Explain how half-life of radioactive element can be determined. OR A 50.0 g sample of carbon is taken from the pelvis bone of a skeleton and is found to have a carbon-14 decay rate of 200.0 decays/min. It is known that carbon from a living organism has a decay rate of 15.0 decays/min-g and that C-14 has a half-life of 5730 years = 3.01×10^9 min. Find the age of the skeletal system.
- xi. Explain any two consequences of Special Theory of Relativity. OR What are inertial and non-inertial frames of references? One example of each is required.
- xii. An FM radiotransmitter has a power output of 100kW and operates at a frequency of 94MHz, How many photons per second does the transmitter emit? OR Cut-off frequency for the photo electric effect in some materials is $8 \times 10^{13} \text{ Hz}$. When the incident light has a frequency of $1.2 \times 10^{14} \text{ Hz}$, the stopping potential is measured as -0.16V. Estimate a value of Planck's constant from these data and determine the percentage error of your estimation.
- xiii. Differentiate ferromagnetic, diamagnetic and paramagnetic substances. (any 2) OR A farmer making juice fills a glass bottle to the brim and caps it tightly. The juice expands more than the glass when it warms up in such a way that the volume increases by 0.2% (i.e., $\frac{\Delta V}{V} = 2 \times 10^{-3}$) relative to the space available. Calculate the normal force exerted by the juice per square centimeter if its bulk's modulus is $1.8 \times 10^9 \text{ N/m}^2$. Assuming that the bottle does not break.

SECTION C

(DETAILED ANSWER QUESTIONS)

(32 Marks)

NOTE: Attempt any Two questions from this section. All questions carry equal marks. Draw diagram where necessary.

- Q3.(a) What is LASER? Define the meta-stable state, Optical Pumping & population inversion. Describe He-Ne laser in detail. OR Describe the J.J. Thomson method to determine the charge to mass ratio of an electron.
- (b) What is heat engine? Describe the working of the Carnot engine and derive its equation for the efficiency? OR Show that the pressure equation is $P = \frac{1}{3} \rho v^2$.
- Q4.(a) What is a transformer? On what principle does it work? What is the difference between a Step Up and Step down Transformer? Why is a soft iron core used in a transformer? OR What is mass spectrograph? Write down its construction and working. Derive its relevant mathematical expression.
- (b) State Faraday's Law of Electromagnetic Induction. Explain the Phenomenon of Mutual Induction Or Self Induction? OR State Ampere's Law. Derive expression for magnetic field of solenoid.
- Q5. (a) What is Photoelectric Effect? What remarkable experimental results are obtained about the photoelectric emission? Give Einstein's explanation of the photoelectric effect on the basis of quantum theory of radiation. OR Describe the phenomenon of Compton's Effect. In Compton's scattering process, the fractional change in wavelength of X-ray photon is 1 % at an angle of 120° . Find the wavelength of X-rays used in this experiment?
- (b) State first law of thermodynamics. By using first law of the thermodynamics explain Isothermal and Adiabatic process. OR Derive an expression for energy of an electron in hydrogen atom.